A PROPER SURJECTIVE MORPHISM OF LOCALLY QUASI-COMPACT SPACES IS A QC COVERING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the finite covering of the fibre, the closed set Z and the final inclusion are all inside the counted span.

Editorial note. The label lemma-proper-surjective-is-qc-covering occurs in no other staged Stacks chapter, so no cross-chapter restatement had to be weighed for this card.

Editorial note. The single labelled reference occurring inside the counted spans points to a chapter of the Project not staged in this card's source tree; it is carried in assembly.external_references as a non-load-bearing navigation pointer recorded at the invoking line.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The category LC of Hausdorff locally quasi-compact spaces and its quasi-compact subsets. Same chapter, the category LC and the notion of a quasi-compact subset used in the definition of qc coverings. Supplies the ambient category and the quasi-compact neighbourhoods chosen in the proof.
- qc covering. Same chapter, the definition of a qc covering as a family such that every point has a neighbourhood covered by finitely many quasi-compact subsets of the sources. Supplies the property verified for the single morphism f.
- Properness: closedness and quasi-compactness of geometric fibres. Topology, the characterisation of proper maps by closedness together with quasi-compactness of the fibres. Supplies both the closedness of f and the quasi-compactness of the fibre X_y that the proof uses.

Source reference. Original theorem 005R: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7599–7599. Cross-chapter navigation pointer invoked at proof line 7599, where the author uses that a proper map is closed. The target chapter is not staged in this card's source tree, so the pointer records the invoking line; it only resolves the reference for the reader, and no bibliography citation is involved.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of LC. If f is proper and surjective, then $\{f : X \to Y\}$ is a qc covering.*

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Proof. Let $y \in Y$ be a point. For each $x \in X_y$ choose a quasi-compact neighbourhood $E_x \subset X$. Choose $x \in U_x \subset E_x$ open. Since f is proper the fibre X_y is quasi-compact and we find $x_1, \ldots, x_n \in X_y$ such that $X_y \subset U_{x_1} \cup \ldots \cup U_{x_n}$. We claim that $f(E_{x_1}) \cup \ldots \cup f(E_{x_n})$ is a neighbourhood of y. Namely, as f is closed (Topology, Theorem 005R) we see that $Z = f(X \setminus U_{x_1} \cup \ldots \cup U_{x_n})$ is a closed subset of Y not containing y. As f is surjective we see that $Y \setminus Z$ is contained in $f(E_{x_1}) \cup \ldots \cup f(E_{x_n})$ as desired. $\square$